

Surname

Forename(s)

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

5008/2

PAPER 2 Theory

JUNE 2017 SESSION

2 hours

Candidates answer on the question paper

Additional materials: Electronic calculator
Pencil (type HB is recommended)

Allow candidates 5 minutes to count pages before the examination.

This booklet should not be punched or stapled and pages should not be removed.

TIME 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page and your centre number and candidate number on the top right corner of every page of this paper.

Check if the booklet has all the pages and ask the invigilator for a replacement if there are duplicate or missing pages.

Section A

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B

Answer any **three** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question. You are advised to spend no longer than 45 minutes on **Section A** and 1 hour 15 minutes on **Section B**.

This question paper consists of 18 printed pages.

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Section A

Answer all questions in this section.

You are advised to spend no longer than 45 minutes on this section.

- 1 (a) Fig. 1.1 shows a cross section of a leaf.

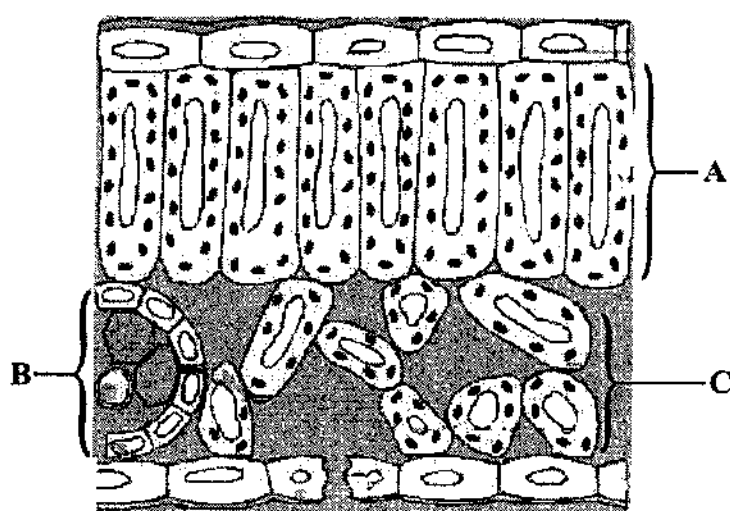


Fig. 1.1

- (i) Name tissues A and B.

A _____

B _____

[2]

- (ii) State any **two** visible differences between cells in tissues labelled A and C.

1. _____

2. _____

[2]

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- 1 (b) Explain any **four** ways by which productivity in a green house can be increased.

1. _____

2. _____

3. _____

4. _____

[4]
[Total: 8]

- 2 (a) State any **three** advantages of cross-pollination.

1. _____

2. _____

3. _____

[3]

4

- 2 (b) Fig. 2.1 shows the structure of a flower.

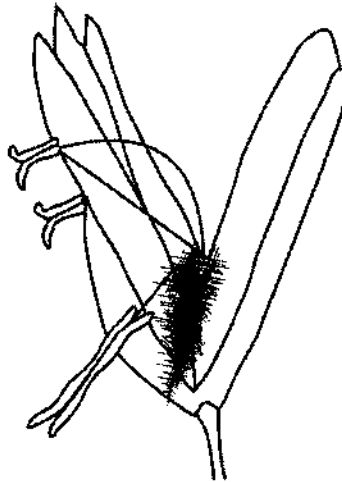


Fig. 2.1

State, with reasons the type of pollination in the flower.

type of pollination _____

reason:

1. _____
2. _____

[3]

- (c) Suggest any **two** features of wind dispersed seeds.

1. _____
2. _____

[2]

[Total: 8]

5

- 3 (a) Fig. 3.1 shows a food chain.

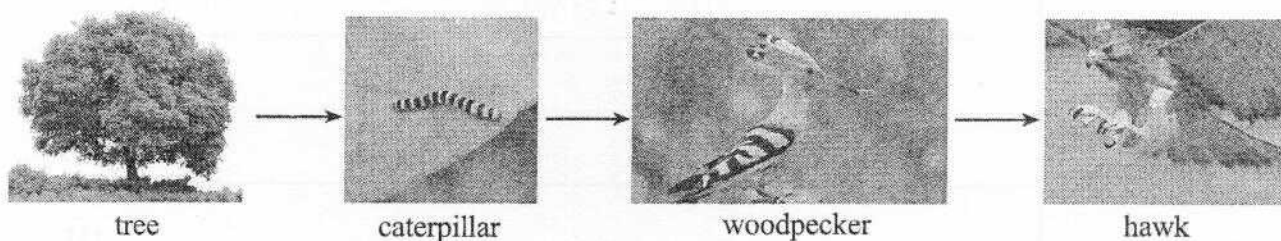


Fig. 3.1(a)

The food chain is represented by two types of pyramids, X and Y.

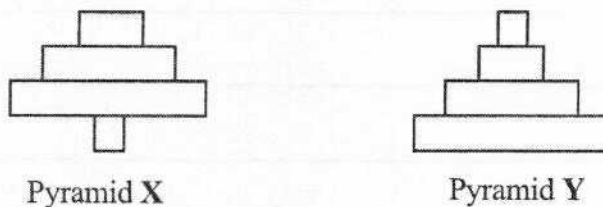


Fig. 3.1(b)

- (i) Name the type of pyramid represented by

X, _____

Y. _____

[2]

- (ii) Explain the shape of pyramid Y.

[2]

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- 3 (b) Explain how legumes improve soil fertility.

[2]

- (c) Explain how leaching of nitrate fertilisers may cause death of fish in rivers.

[2]

[Total: 8]

7

- 4 (a) Define the term *mutation*.

[1]

- (b) Fig. 4.1 illustrates the inheritance of horns in cattle.

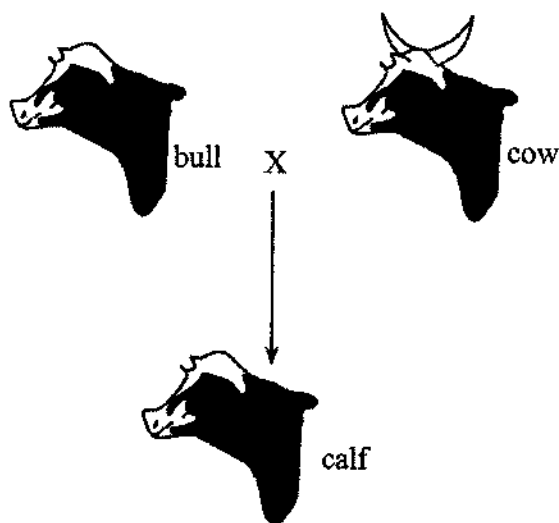


Fig. 4.1

- (i) Deduce the phenotype of the dominant allele.

[1]

- (ii) Using symbols **H** and **h**, state the genotypes of the bull, cow and calf.

1. bull,

2. cow,

3. calf,

[3]

- (c) Explain how sex is determined in mammals.

[3]

[Total: 8]

8

5 (a) State any **three** functions of the skeleton.

1. _____
2. _____
3. _____

[3]

(b) Fig. 5.1 shows the bones and muscles of the arm.

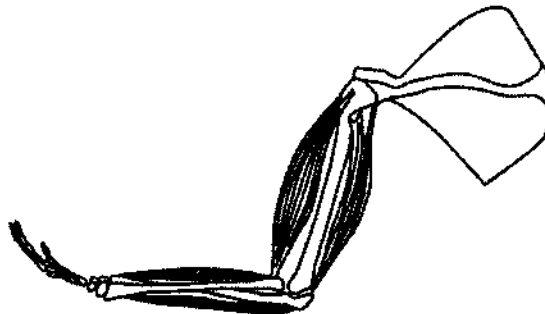


Fig. 5.1

(i) Name the type of joint at the elbow.

_____ [1]

(ii) Label, on Fig. 5.1, the triceps muscle.

[1]

(iii) Describe what happens to the muscles when the arm straightens.

_____ [3]

[Total: 8]

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Section B

*Answer any **three** questions from this section.*

You are advised to spend no longer than 1 hour 15 minutes on this section.

- 6 (a) Explain why a particular enzyme only acts on a specific substrate.

[4]

- (b) Explain the effect of each of the following on the rate of an enzyme catalysed reaction:

- (i) reduction in temperature

[2]

- (ii) changing pH

[2]

- (c) Explain the importance of digestion.

[5]

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[5]

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- 7 (b) (i) Explain the role of arterioles and sweat glands when a person is subjected to cold conditions:

arterioles

sweat glands

[6]

- (ii) Explain why blood flow to the skin increases during exercise.

[3]

- (c) (i) State any **four** signs and/or symptoms of diabetes mellitus.

[4]

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- 7 (c) (ii) Describe how diabetes mellitus is controlled.

[3]

[Total: 20]

- 8 (a) Explain why a plant wilts on a dry and sunny day.

[5]

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- 8 (b) Describe and explain how xylem vessels are adapted for transport of water, and for support in a plant.

[4]

- (c) Describe how the platelets, phagocytes and lymphocytes help to protect the body against infection.

platelets _____

phagocytes _____

lymphocytes _____

[6]

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- 8 (d) Outline how active immunity is acquired.

[5]

[Total: 20]

- 9 (a) State any **four** characteristics of fungi.

1.

2.

3.

4.

[4]

- (b) (i) Describe how cholera is spread.

[4]

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- 9 (b) (ii) State and explain any **three** methods of reducing the spread of cholera.

method 1 _____

explanation _____

method 2 _____

explanation _____

method 3 _____

explanation _____

[6]

- (c) (i) Name the causative agent and vector of malaria.

[2]

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- 9 (c) (ii) Suggest any **four** reasons why it is difficult to completely eradicate malaria in Zimbabwe.

[4]

[Total: 20]

- 10 (a) Explain how each of the following hormones control the menstrual cycle:

- (i) oestrogen

[3]

- (ii) progesterone

[3]

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- 10 (b)** State any **four** methods of contraception.

[4]

- (c)** Describe the effects on an unborn baby of

- (i)** drinking alcohol,

[3]

- (ii)** smoking tobacco.

[3]

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- 10 (d) Explain any **two** ways an HIV positive individual may prolong life.

[4]

[Total: 20]